

Ex No.9 - Massive MIMO Capacity vs SNR and Gain Using MATLAB

Objective – 1 Plot of Channel Capacity (bps/Hz) versus SNR (dB).

MATLAB Code:

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

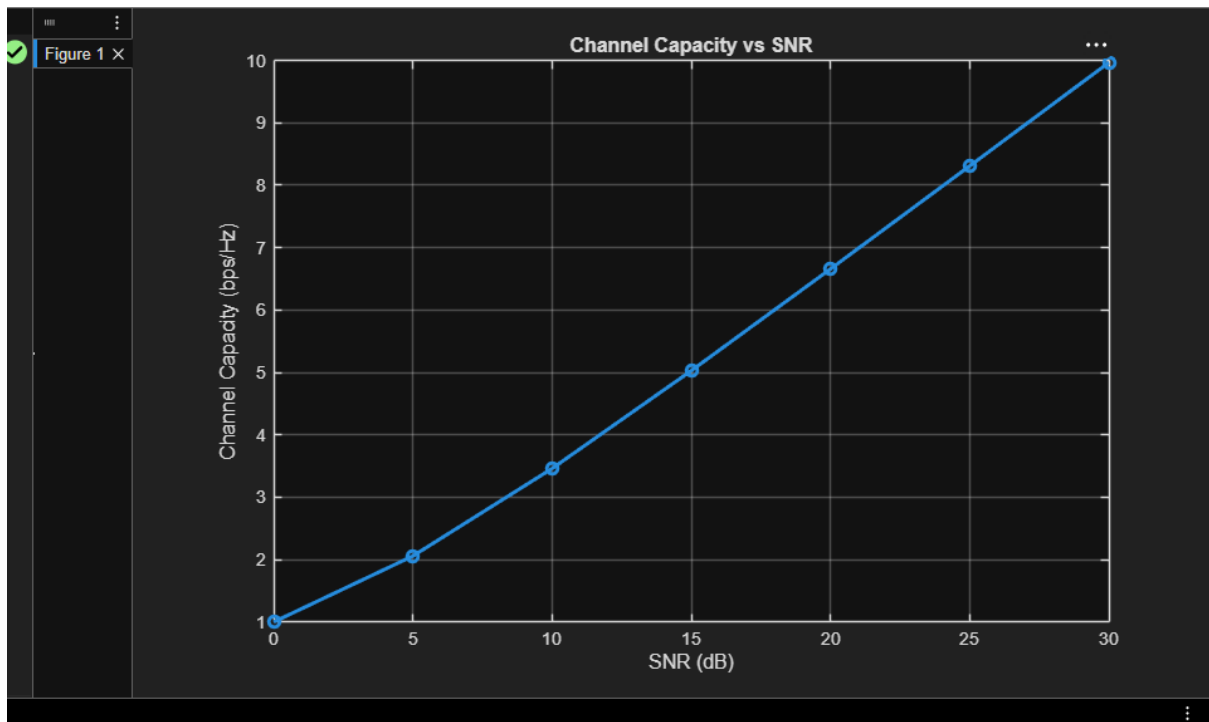
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```



Objective – 2 Bar graph comparing SNR values under different communication scenarios.

MATLAB Code:

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

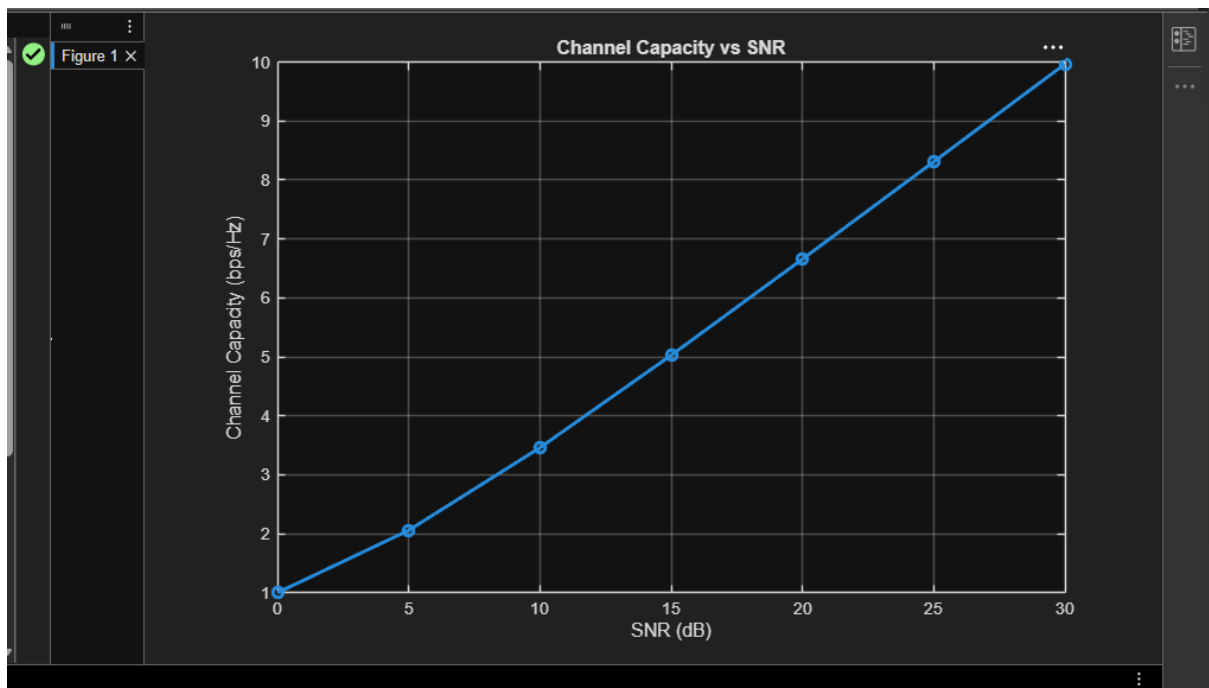
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```



Objective – 3 Bar graph showing Antenna Gain for different antenna array sizes.

MATLAB Code:

```
clc;

clear;

close all;

antennas = [4 8 16 32 64];

gain = 10*log10(antennas);

figure

bar(antennas, gain)

grid on

% Number of Antenna Elements

% Antenna Gain (dB)

xlabel('Number of Antenna Elements')

ylabel('Antenna Gain (dB)')

title('Antenna Gain for Different Antenna Array Sizes')
```

